Classical ECDLP experiment programme

25 October 2026. Prospective designs, mathematical controls and implementation checks; **no admitted scientific trials or attack advantage**. This follows the research plan. Classical CPU/GPU algorithms only. Factor-base relation collection, index calculus, quantum algorithms and quantum simulation are outside this programme.

Coverage and purpose

Cover the direct tagged-response proposal and all nine classical lead families named in the plan. LEADS.md gives a separate card for each family. This is exhaustive coverage of that named list, not of the literature, every repository branch, or all possible algorithms. An experiment can exhaust a finite, frozen search grammar; it cannot exhaust an unspecified mathematical idea.

The desired research outcome remains a constructive, previously untested EC-specific algorithm. Its novelty and complexity are unverified. Collecting reproducible ordinary, unusual, negative or inconclusive observations is also a valid study outcome. Correct execution, mathematical falsification and competitive performance are separate questions.

Several originals, including IDEA-050, IDEA-111, IDEA-156 and IDEA-379, were formulated for factor-base/index-calculus pipelines. Their original pipelines are excluded. Their cards describe explicitly different direct-inversion successors whose missing constructions remain visible. A proved obstruction is a control to reproduce and respect; it is not a reason to rerun the same algorithm under a new name.

Outcome vocabulary

Each trial reports one execution outcome: `complete_valid`, `wrong_output`, `timeout`, `memory_limit`, `invalid_input`, `infrastructure_error`, or `not_executable`. Preserve every attempt, including failed searches and constructor failures. A counterexample falsifies its exact universal identity; a timeout does not. No outcome automatically changes an official lead or hypothesis status.

Distinguish five research milestones:

1. **Encoding correctness:** exact agreement with an independent reference, including exceptional cases and unique witnesses. This qualifies the instrument only.
2. **Observation:** a frozen statistic with its inputs, reference, uncertainty and all outcomes. An anomaly is unnecessary.
3. **Mechanism correctness:** an explicit public constructor, an identity or invariant with exact domain, and closure under composition. One counterexample rejects that version.
4. **Scoped performance:** measured complete costs against matched baselines. A local reduction is labeled local; it is not a solver gain.
5. **Novel algorithm or asymptotic claim:** additionally requires prior-art collision review, independent reconstruction, a uniform algorithm and a justified time/memory analysis. A short size ladder cannot establish an exponent.

Shared input and blinding contract

Development fixtures already implemented are listed in CHECKS.md. They are tiny synthetic unit-test inputs, not the registered scientific cohort below.

Prospective discovery cohort: subgroup bit lengths 8, 10 and 12; four curves per length; exhaustive canonical scalars. **Prospective confirmation cohort:** six fresh curves at each of 12, 14, 16 and 18 bits, eight independently drawn uniform scalars per curve: 24 curves and 192 random targets. Keep the forced scalars 0, 1 and $r-1$ and exceptional arithmetic cases in a separate correctness suite; do not replace random performance targets with them. The two cohorts have disjoint curve IDs and generator domains. These exact scientific inputs have not yet been generated or registered.

Before admission, register each (p,a,b,#E,r,h,P), canonical point encoding, subgroup and field certificates, curve-family stratum, generation seed/algorithm, rejection history and content digest. For

these sizes, exact point enumeration can certify #E, trial division can certify primality, and [r]P=O with P≠O certifies the prime subgroup order. Check nonsingularity, odd p, r prime, r≠p, #E≠p and exact cofactor h=#E/r. Specify the public registry authority; do not silently invent canonical IDs.

Freeze a deterministic hash-stream rejection sampler and its accepted/rejected records, rather than choosing curves after inspecting results. The main stratum is ordinary non-anomalous curves with j different from 0 and 1728; special-j, special-prime, supersingular or extension-field families require separately registered strata and matched baselines. Describe the induced sampling distribution; it is not uniform over all elliptic curves. Include a nontrivial-cofactor fixture and targets outside the subgroup as rejection/zero-response controls.

Freeze and publish the constructor, compiler options, grammar, elimination schedule policy, stopping rule and input-manifest hashes before opening confirmation targets. An input custodian generates synthetic public Q values and keeps k, scalar-derived filenames and reverse lookup tables outside the candidate process. An output checker alone uses the truth. Record custody and process access. A public seed that reconstructs k is not a blind target. If isolation is unavailable, call the test non-blind. A new method version gets a fresh confirmation cohort; do not retune on its held-out results.

Curves are the replication units. Eight targets on one curve share setup and arithmetic and are not eight independent curves. Report per-curve summaries and every target. Use a curve-cluster bootstrap for descriptive uncertainty in paired cost ratios; with six curves per size, call uncertainty coarse. These counts are an initial diagnostic allocation, not a power calculation for a particular conjecture. Rare-event hypotheses require a separately frozen sample-size calculation from their predicted probability.

Direct response: an executable specification and four tests

For canonical 0≤b<r, total scalar multiplication C(b) and target Q, use

$$R = \mathbb{F}_2[\varepsilon_0, \dots, \varepsilon_{\ell-1}]/(\varepsilon_i\varepsilon_j \text{ for all } i,j), \qquad Z_Q = \sum_b [C(b) = Q] \left(1 + \sum_j b_j \varepsilon_j\right).$$

For Q=[k]P there is one accepted scalar and Z_Q=(1,k). A square-zero ring element is an ordinary pair (constant bit, tag bit-vector); addition is XOR and (a,u)(b,v)=(ab,av+bu). This is known algebraic model counting. Enumerating all b costs O(r) scalar evaluations and is a reference, never the proposed speedup.

D0 — instrument checks. Enumerate all bit inputs, including b≥r, all subgroup targets and selected outside targets on fixed tiny fixtures. Compare independently coded affine and Jacobian arithmetic, enumerate exceptional additions, and count satisfying transition witnesses. Inject one unconstrained Boolean auxiliary variable and verify that two identical witnesses cancel in characteristic two. Inject an intentionally wrong rewrite and ensure the checker returns a boundary counterexample. Check tagged coefficients, not just ordinary parity. Pass: zero mismatches in the declared finite suite and successful detection of each injected defect. Failure: a concrete mismatch invalidates that encoding/checker version. Limits: point-level deterministic arithmetic is not a gate-level Boolean circuit or a carry/wire proof.

D1 — exact micro-block discovery. First construct total, gate-level one- and two-step double-and-add blocks, b<r and canonical-output constraints. Fix wire layout, finite rewrite grammar, coefficient domain and an exact boundary enumeration before search. Candidate output must name the original block, replacement, target pins, every retained boundary bit and construction cost. Compare reference contraction to replacement for every boundary assignment on registered tiny instances, then overlapping two-block compositions. Require equality of all coefficients or a proved narrower invariant closed under composition. Generic XOR elimination, Boolean simplification and ordinary fixed-window merging are controls. Preserve every rejected candidate and its first counterexample. A finite grammar can be exhausted only after its file and bounds are frozen; that file and a gate compiler are currently missing.

For a compiled-counting adapter, test decomposable conjunctions, deterministic disjunctions and smoothing explicitly. Positive and negative scalar-literal weights sum to epsilon_j, not to 1. Omitting an unused scalar variable therefore changes the tagged count unless its actual weight sum is included. Verify the chosen compiler's theorem and coefficient-domain assumptions rather than substituting ring operations into an arbitrary Boolean DAG.

D2 — structure and construction measurement. For each surviving rule measure constraints eliminated, maximum participating-index union, peak live entries, encoded coefficient bits, constructor operations, rule applications and total checking/recovery work. Run the same generic simplifier on the EC block, an injective affine-bit-map control and a tiny explicit random-permutation control. Charge permutation table generation/storage. Arithmetic-mutated controls must retain total unique-extension semantics and disclose their encoding differences. An invertible basis change alone preserves the original function and is not a destroyed-structure null. Record whether any difference is specifically attributable to the EC identity. A small answer, separator, numerical rank or expression is insufficient if its construction is expensive.

The fixed-window matrix control in window-control.md has scalar rank $r-1$ over F_2. It explains why a dense global point-state low-rank replacement cannot be presumed. It does not prohibit a different bit-level, nonlinear or target-conditioned algorithm. No gate-level discovery or D2 comparison has run.

D3 — frozen complete solver. Admit only after there is an actual constructor and composition argument. Run every random confirmation target and the separate correctness suite. Recover k using public inputs alone and verify [k]P=Q independently. Charge failed branches, compilation, schedule search, target specialization, contraction, candidate extraction and verification. Compare a cold first-target run with matched rho, relevant automorphisms, BSGS where memory permits, and the strongest available exact compiler on the identical problem. If a comparator is unavailable, state that limitation and do not claim superiority to it. A generic integer #SAT result is not automatically a characteristic-two tagged count; bind and verify the adapter.

For an exploratory local-gain screen, a twofold reduction in the declared dominant local operation on two registered sizes warrants a fresh composition/cost test. That threshold is an engineering choice, not a theorem or a publication rule. For a complete-solver performance study, freeze its minimum effect and paired estimator only after baseline calibration; include A/A repeatability first. No performance threshold is retrofitted to results.

Baselines, costs and statistics

Use identical public Q values, curve strata, arithmetic backend, threads, compiler, hardware placement and stopping/correctness policy. Freeze baseline tuning on development inputs. Rho uses negation and applicable automorphisms with cycle handling; credit known endomorphism speedups to both sides where applicable. A BSGS comparator gets its actual storage budget. Record baseline seeds and stochastic spread; do not compare a lucky candidate run with a single unlucky walk.

Use A/A repetitions to expose slot/host drift, randomize candidate/baseline execution order and record process placement and frequency/thermal conditions when available. Wall time, CPU time, field additions/multiplications/inversions, bit operations and allocation/storage bytes have distinct units. Raw XOR counts are not directly comparable to group additions. A full-cost elapsed-time claim needs controlled calibration and placement. Preserve compilation cost and circuit-size/width counters even when timing noise dominates.

Primary per-target cost is cold setup + preprocessing + target work + failed attempts + recovery + verification; record peak live and persistent memory separately. Secondary many-target costs must name the reuse count m and report (setup+sum target costs)/m. Compare with preprocessing baselines for the same m. Do not hide an O(r) truth table in setup. Preserve censored/time-limited targets, success fraction and resources consumed; successful-only averages are prohibited.

Report descriptive distributions and paired per-curve ratios. Any exploratory size fit states its model, range, confidence and dependence assumptions; it cannot establish an asymptotic exponent. Confirmation of a new speed/attack-cost claim requires repository progress-fidelity records and an independent session, neither supplied by this design document or its assisting agents.

Frozen receipt and machine protection

Every scientific run needs immutable code/commit and dirty state, input and protocol hashes, exact argv, compiler/build identity, host/resource identity, UTC times, seeds, outcome, raw counters with units, output/certificate hashes, checker identity and all stdout/stderr. Record requested/completed/failed/censored counts by arm. Archive a failure before retrying; link the new attempt. Large artifacts follow the shared object-store policy, with hashes and readback; do not silently drop data to fit Git limits.

Use one worker initially and an explicit RAM limit below the host allocation. Set a process watchdog from the declared worst-case input and a resource estimate during admission. These protect the host and classify censoring; they are not lifetime research budgets or evidence of impossibility. The programme has no arbitrary campaign spending cap. Fixed cohorts end their registered test; subsequent work uses an additive new protocol.

Readiness and next actions

The repository requires a prospective runnable specification, launcher, committed input/source hashes, executor handoff/queue and lawful lane before approval. Its read-only `experiment_preapproval.py` must return `approvable: true`; after the committed approval/archive, the selector must return `execution_state: ready` before scientific dispatch. No such receipt is fabricated here. Development self-tests below exercise code, not those admission gates.

Immediate work completed: per-lead designs, a point-level C++ reference, finite correctness checks and a scoped window-control derivation. Immediate implementation debt: registered scientific cohorts; a bit-level total circuit and unique-wire checker; exact finite grammar; lead-specific constructors and adapters; complete launch/admission/custody records. See READINESS.json and the named obstruction.

Prioritize D0 instrument completion and D1 exact blocks. In parallel, reproduce the lossy-panel finite references and deterministic rigidity proof as controls, and test only explicitly supplied EDS/endomorphism/automaton constructions. Do not run undefined coefficient, inverse-state or Pfaffian oracles. New direct successors of blocked originals get fresh identities rather than rewriting frozen evidence. Revisit a scoped negative only when its stated assumption changes or a specific missing constructor appears.

This programme changes no canonical frontier or official scientific disposition. It provides a concrete path to distinguish a correct instrument, informative observation, failed construction, resource limit, local identity and complete classical algorithm.

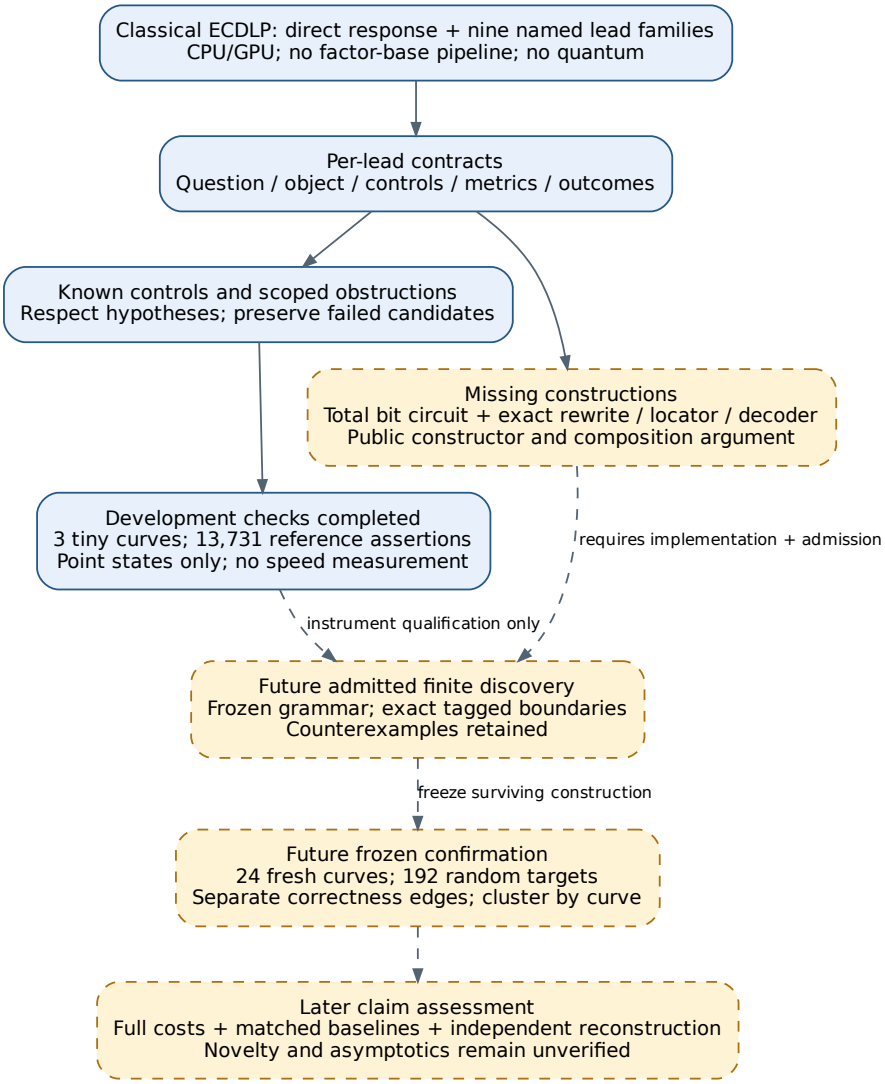


Figure 1: Completed development scaffolding and proposed scientific stages. Dashed amber boxes are pending; no measured speedup is depicted.

Experimental cards for the nine cited classical lead families

Design date: 2026-10-25. These are proposed protocols, not approved experiment records, executed trials, or scientific dispositions. They exhaust the nine named prior-family rows in the companion plan, not classical ECDLP research. The direct tagged-count workstream has its own card. No card authorizes quantum computation or index calculus.

Common design and outcome vocabulary

Use public synthetic non-anomalous odd-prime-field instances with a certified odd-prime subgroup order `r`. Register the companion programme's proposed discovery fixtures before scientific admission: four curves at each of 8, 10 and 12 subgroup bits. All canonical scalars are the discovery truth set; a truth process may enumerate them, but its tables and labels are not candidate inputs. Freeze the algorithm, feature definitions, rational/finite-field coefficient domain, exceptions, seeds, and exact manifest before fresh tests. A candidate that survives discovery uses the companion held-out allocation of six fresh curves at each of 12, 14, 16 and 18 bits and eight targets per curve. This is a contingent allocation, not existing fixtures or an asymptotic argument. Special automorphism families are separate strata with matched baselines.

Every run records exact successes, misses, false outputs, multiplicities, undefined cases, operation counts, setup, discarded trials, peak live storage, serialized state/coefficient sizes, and verification. All cold construction costs count. Discovery enumeration is instrumentation, not a solver speedup. A theorem-only card records a proof obligation rather than running a benchmark with an assumed oracle. Resource limits censor trials; they do not refute mathematics.

Three different outcomes must be retained: **measurement success** means the declared observations were obtained and controls passed; **candidate failure** means an exact claimed property has a counterexample or a measured cost objective failed at the tested scope; **inconclusive/impeded** means missing construction, inadequate resolution, failed controls, or infrastructure. Ordinary behavior and null observations are complete useful measurements. A mechanism signal or stage gain is separate from a whole-solver gain. No empirical fit on this ladder proves an exponent below one half.

Each candidate implementation gets a collision paragraph against its cited record and the generic tool that already performs the operation. Published interpretation requires independent checking under the repository contract. These cards do not reopen retired hypotheses or modify frozen gates.

L1. Prime-to-p lifts and non-homomorphic local sections

Prior boundary. `LEAD-EXO-padic-jet-canonical-lift` records the prime-to-characteristic additive-log obstruction and previous hard-lift versus computable-section tests. An additive homomorphism from an order-`r` group into a p-primary additive target is zero when `r != p`. That scoped theorem needs no performance rerun. A hard lift defined as `[k]S(P)` already contains the unknown scalar. The potentially testable successor is a specified, cheap, non-homomorphic section; it does not inherit additive linearity.

Concrete diagnostic. For a nonsingular integral lift of `E` modulo `p^2`, freeze `A,B` and an explicit point section `S`. One reproducible candidate chooses the Teichmuller lift of the canonical x-coordinate and Hensel-lifts the observed y-coordinate; handle infinity separately. For an odd-order subgroup, a nonidentity rational point has `y != 0`, so the y derivative is a unit. Compute `D_S(R) = t([r]S(R))/p mod p`, with `t=-x/y` at infinity using a correct projective implementation. Record the defect `D_S(U+V)-D_S(U)-D_S(V)` and the entire canonical-scalar profile `D_S([k]P)`. These are computable from public points without k. Other section families must be declared separately before measurements; do not silently tune a section after observing labels.

Measurements and controls. On discovery curves enumerate all k and all pairs on the smallest four curves; use a fixed 65,536-pair sample on the remaining curves if exact pair enumeration is not selected in the frozen plan. Positive controls are the genuinely additive formal-group law and a known anomalous-curve control kept outside the target population. A deliberately hard lift `[k]S(P)` is a labeled leakage trap whose linearity is expected, not an attack. Compare the same fixed scalar

predictor and profile statistics with 99 permutations of scalar labels preserving the feature multiset. Record the sign symmetry and identity separately. Report the exact chance reference induced by each permutation scheme rather than asserting a universal `1/p` accuracy.

Advance criterion. A new section has a reproduced public-computable relation or frozen predictor whose held-out deviation survives controls; report its effect and uncertainty before interpreting it. Exact scalar recovery then requires an explicit decoder, complete candidate list, and charged costs. **Failure:** one counterexample rejects claimed exact additivity; hard-lift advice rejects the claimed public constructor. Matching null labels is a scoped observational null, not closure of every section. **Impeded:** no explicit section, insufficient test resolution, or undefined exceptional lifts. The next mathematical question after a signal is why the section defect preserves scalar information across unseen curves and wraparound.

L2. Elliptic nets / EDS and target-specific nonlinear locators

Prior boundary. `LEAD-EXO-elliptic-net-eds` and IDEA-006 already cover short linear annihilators, translated coordinate blocks, ordinary displacement-rank tautologies, and the old Somos sieve. Repeating that sieve is outside this programme because it is a relation-collection route. Net recurrences do not supply an index inverse. The source itself leaves target-specific nonlinear locators and an unexplained rank-two-net null asymmetry open; obtain the raw prior evidence before treating that asymmetry as a fresh observation.

Runnable diagnostic. Freeze exact division-polynomial/net recurrences with normalization and zero handling, and build (i) known-index net values and (ii) target-translated public coordinate blocks `a_Q(n)=x(Q+[n]P)` with a distinct infinity symbol. Verify every recurrence on discovery truth, including zeros. Measure minimal linear complexity of every prefix of lengths `8,16,...` up to the available period, both point signs, and nonlinear-predictor residuals for a frozen finite grammar. A permissible exploratory grammar is quadratic polynomials in a four-value coordinate window, fitted on disjoint windows and tested on all untouched windows; rank/identifiability and coefficient-storage costs are reported. This measures prediction, not scalar location. Arithmetic domains and infinity charts must be explicit.

For that finite quadratic grammar there are exactly 15 monomials of degree at most two in four variables. Fit a deterministic linear system for `a(n+4)` over `F_p`; freeze lexicographic monomial order and a canonical free-variable assignment. Use the first half of finite-chart windows for fitting and the second half for testing, and report all infinity-crossing windows separately. Excluding a window is permitted for this declared chart diagnostic only; a solver still has to handle it. A second grammar is a separate registered search, not silent post-hoc tuning.

Controls. Same-normalization known EDS identities; matched random periodic sequences; random scalar relabeling of the exact same coordinate/net-value multiset; Berlekamp-Massey and the existing recurrence as predictor controls. A hidden scalar shift in the training labels is a leakage trap. A low-degree recurrence that predicts values while leaving all cyclic orientations feasible is recorded as such.

Constructor gate. Before any direct-recovery benchmark, supply an explicit locator whose coefficients/representation are constructed from `(E,P,Q)` and whose roots or reverse path yield the canonical k, with a charged uniform construction and complete ambiguity list. Name the algebraic identity beyond the already-known recurrence. A sparse expression naming an unknown index is not a constructor. **Success:** exact recurrence/locator measurements with all controls; a useful candidate additionally reduces orientation ambiguity on fresh targets with charged total cost. **Failure:** one false locator root/missed scalar rejects exactness; linear complexity above a claimed bound rejects that bound on its declared family. **Null:** frozen nonlinear prediction matches its controls. **Impeded:** the nonlinear locator has not been supplied. No finite predictor panel closes all nonlinear locators.

L3. Endomorphism actions and coordinate observables

Prior boundary. `LEAD-STR-endo-scalar-action-generic` covers group-element oracles with known eigenvalues: on the prime-order subgroup, `phi_i(R)=[lambda_i]R`, so `phi_i(Q)=[k]phi_i(P)` is the

same unknown congruence. It does not rule out every computation on actual field-coordinate bits. Known constant-factor automorphism gains remain legitimate matched controls.

Exact control experiment. In each declared automorphism stratum, verify `phi_i(P)=[lambda_i]P` and `phi_i([k]P)=[lambda_i k]P` for every discovery k. For every proposed short witness-lattice vector c, classify the certificate: `sum c_i lambda_i = 0 mod r` is k-independent, while any claimed scalar constraint must be written explicitly and checked against all k. The `j=0` relation `1+lambda+lambda^2=0` is a deliberate positive trap: a short vector satisfying it cannot recover k. Compare any generic walk with the same walk using the same automorphism quotient, not a baseline deprived of the quotient.

Successor gate. A new candidate must name an efficiently computable field-valued or bit-valued observable and the exact operation it supplies beyond group-image actions. Freeze its decoder and candidate ambiguity before held-out tests. Do not accept an unimplemented "coordinate oracle." **Success:** complete exact certificate classification and measured constant-factor effects, or a separately constructed coordinate operation that survives blind recovery. **Failure:** k-independent vectors reject the claimed scalar inference; scalar-indexed eigenvalue/orientation tables reject the constructor. **Null:** no additional recovery beyond the matched automorphism baseline at tested sizes. **Impeded:** no non-generic observable/decoder has been constructed. The generic theorem is not a blanket lower bound on coordinate algorithms.

L4. Matchgate / Pfaffian transform, direct-inversion successor to IDEA-050

Scope correction. IDEA-050's original hypothesis is explicitly a factor-base relation/matrix/descent scheme and is outside this user's scope. Do not run or relabel its original experiment as non-IC. This card is a different, still-unconstructed direct-inversion successor: transform the exact scalar-program factor network conditioned on Q. Matchgate computation and basis transforms are established tools, not the proposed novelty.

Finite preflight. Freeze one- and two-step total scalar blocks, their boundary variables, equality/fan-out tensors and target pins. Start over exact rational weights so matchgate signs and coefficient bit growth are visible; do not transfer a characteristic-zero Pfaffian theorem to characteristic two or the tag ring without a proof. Candidate basis matrices must be explicit, invertible, public and constructed without k. Evaluate all parity-support and matchgate/Grassmann-Pluecker identities exactly on every local transformed signature. Include membership/range/equality/exception tensors, not just a favorable arithmetic gate. A basis-search grammar and its finite bounds must be committed before search; every failed arm is retained.

One concrete first search grammar is a single common 2-by-2 rational matrix M whose entries are in `{-1,0,1}`, excluding determinant zero. Fix the index convention so one end of each contracted wire transforms by M and the other by its inverse transpose; verify that their contraction is unchanged. There are at most 81 arms before singular matrices are removed. Check the identical M against every required tensor of the frozen block. This very narrow grammar is deliberately exhaustive within its declared bounds; failure says nothing about all bases. Larger rational entries, per-wire bases, extension fields or alternate planar realizations require separately frozen successor grammars.

Composition and controls. Positive control: a known planar matchgate network conjugated by an explicit invertible basis. Negative control: random signatures with identical arity and declared support. Mechanism control: untransformed exact contraction and the same generic simplifier. For a surviving basis, construct the complete graph, account for crossings/planarization and bit tags or repeated scalar-bit conditioning, and verify the Pfaffian answer against direct contraction. A valid local signature does not establish a planar tractable whole network.

Advance criterion. Exhibit the basis, realizations, exact identity checks, global graph construction, and a source-free recovery procedure for every scalar bit. Charge basis discovery, rational numerator/denominator growth, all conditioned Pfaffians, graph width, zero cases and output verification. **Failure:** a violated exact identity rejects that basis on that signature; missing equality/crossing/exception realization rejects the asserted whole-network reduction; one incorrect bit rejects recovery. A finite unsuccessful basis search rejects only its grammar. **Impeded:** no explicit basis/planar construction; this is the current direct successor's status. A full arbitrary-basis impossibility claim would need a separately proved invariant, not a numerical rank plot.

L5. Radical ladder quotient, target-local successor to IDEA-111

Scope correction and theorem gate. IDEA-111 is theorem-deferred and its original factor-base/descent path is out of scope. Its frozen text explicitly requires independent review before solver execution. Keep that gate and all original records unchanged. A direct target-local functional is a different question, not an automatic escape from the rank obstruction.

First discriminating proof. Write the exact complete finite branch ideal, including selectors, field equations, range restriction, exceptional components and quotient map. For a reduced finite algebra over `F_p` whose complete geometric support consists of m distinct rational points, the coordinate-ring dimension is m by the Chinese remainder theorem. A quotient algebra retaining a bijection to all branch points therefore cannot have dimension below their count. If all length-t branch labels remain recoverable, shared syntax does not avoid this dimension/output obligation. Check whether the proposed quotient genuinely changes those hypotheses, rather than measuring a smaller input DAG.

Only after its gate, finite diagnostic. For approved symbolic depths `t=1,...,6`, enumerate the exact valid path set for reference and compare every decoded quotient point and multiplicity. Record number of branches, radical normal-form dimension, coefficient bits, construction cost and decoder cost. Include repeated-output branches with distinct selector labels, zero denominators, infinity, and both signs. Controls: duplicated identical components with a known quotient; disjoint affine branches with direct-product coordinate ring; the ordinary shared-prefix DAG. One lost or invented branch rejects losslessness at that depth.

Direct target-local question. Once Q is pinned and canonical range enforced, the valid scalar graph has one point, so its reduced coordinate ring has dimension one. That is a correctness fact, not a cheap constructor: computing its generator can already reveal k. A successor must give a public target-conditioned invariant closed under composition without constructing or decoding the full branch union. Measure the full construction, specialization and recovery costs; share this experiment with the direct tagged-count workstream rather than counting it as independent evidence.

Success: a checked structural statement at its exact quantifiers, then an explicit target-local construction and verified compositional rule if one is found. **Failure:** a counterexample to losslessness or the proposed dimension bound. **Null:** no compression beyond the declared duplicate control in the tested depth grammar. **Impeded:** missing theorem/independent review or missing target-local constructor; do not launch an assumed quotient benchmark.

L6. Reverse automata, successor conditions for rejected IDEA-070

Prior boundary. IDEA-070 is `rejected_merged`, with no active experiment gate. Small-state net/orbit descriptions and a missing orientation map are occupied routes. For `r` near `p`, base-`p` scalar notation has essentially one digit, so a putative shallow automaton may hide `r` outgoing labels. A small minimal acceptor of a single already-known target word is also vacuous: every single word has a short acceptor if that word is supplied.

Diagnostic without promotion. Freeze a point-computable observable, base (`p` for the original question; base 2 as a separately named successor), digit order, canonical padding and output alphabet. On the discovery truth sequence, construct the exact bounded-depth residual functions and minimize only for measurement; record all states, transitions, transition-label bits, output ambiguity, constructor cost and number of scalars compatible with each target observation. For base 2 residuals use `n -> a(2^e n+j)` on the explicitly declared finite domain, with periodic/canonical boundary convention stated. A bounded-depth residual count is not a proof of the infinite p-kernel size.

Controls. A known small reversible automaton; the same elliptic observable with a random cyclic orientation; random periodic sequences with the same alphabet frequencies; the ordinary recurrence predictor. Train-on-all-k minimization is a labeled diagnostic only, not available preprocessing for free. Verify all reconstructed paths and retain the full residual candidate set, including the +/- ambiguity of x-only observations.

Constructor gate. A successor needs an algebraic public constructor for the state machine and a point-to-accepting-state orientation that do not enumerate the scalar dictionary. State count, outgo-

ing alphabet, transition labels, reverse branching, setup and verification all count. **Success:** exact state/ambiguity curves and a separately constructed blind decoder. **Failure:** one missing path rejects exactness; an orientation table rejects the claimed cheap constructor; a claimed state/ambiguity bound fails on a counterexample. **Null:** compression/prediction occurs but does not reduce target-to-scalar ambiguity. **Impeded:** absent uniform constructor/orientation, currently the substantive gap. Do not reopen IDEA-070 merely by renaming its automaton.

L7. Conditional coefficients, direct-inversion successor to rejected IDEA-156

Scope correction. IDEA-156's factor-base grid and source self-reduction are out of scope. Its missing cheap conditional-coefficient oracle remains directly relevant. No original experiment ran; retain its rejected disposition. Combinatorial Nullstellensatz is established prior art and supplies no free coefficient evaluator.

Exact direct specification. Let `F_Q(b)` be the companion total unique-extension scalar indicator. Over `F_2`, reduce its polynomial modulo `b_i^2-b_i` to a multilinear representative. For every scalar prefix u, the top coefficient in the remaining variables equals the parity of satisfying completions. Since at most one canonical scalar exists, that parity is an exact 0/1 prefix-existence answer. Thus prefix conditioning can recover k if these coefficients are computed; this is the same missing evaluation problem as the tagged-count card, not a second algorithm.

Finite diagnostic and controls. For every prefix of every discovery target, compute the coefficient by explicit truth summation and independently by Boolean interpolation/finite differences. Record degree, monomial support, coefficient storage and operations, with every auxiliary wire fixed uniquely. Test planted affine Boolean inverse maps, a tiny explicit random permutation and raw zero-polynomial orientation traps. A shared Boolean DAG with OR/AND naively replaced by ring operations is a deliberately invalid control. Charge the truth grid and interpolation; neither is a speed claim.

Advance criterion. Supply an explicit exact coefficient recurrence or compressed construction from `(E,P,Q,u)` with a compositional proof and complete cost. Reuse the direct-core held-out tests and do not double-count their trials. **Failure:** one wrong prefix answer rejects an exact evaluator; even multiplicity canceling a true scalar rejects the encoding; full truth-table interpolation rejects a sub-enumerative claim for that implementation. **Null:** known generic coefficient methods reproduce enumeration-level cost. **Impeded:** no cheap coefficient constructor. Smaller formal degree alone does not bound coefficient construction or density, and a bounded failed grammar does not close all evaluators.

L8. Ax-Katz counts, direct singleton diagnostic for rejected IDEA-379

Prior boundary. IDEA-379's retired zero-run source-locator protocol cannot be relaunched; its restricted-deck descent is also outside scope. Divisibility of a many-point count does not distinguish zero from a positive multiple of p. The singleton direct encoding changes the count range, but it does not provide the count.

Smallest exact control pair. Over `F_p`, the one-variable systems `1=0` (zero solutions) and `0=0` (p solutions) share the receipt "count divisible by p". A divisibility-only existence rule must fail on this pair. Verify the same ambiguity on any concrete count-receipt format proposed by a successor, keeping guaranteed lower bounds on valuation separate from an exact count or exact valuation. The exact valuation of zero is infinity; a theorem's lower bound is not that exact valuation.

Direct diagnostic. Build the total Boolean scalar circuit as polynomial constraints over `F_p`, including `z(z-1)=0`, gate equations, canonical range, target pins and prefixes. Because the extension is unique, the number of solutions is 0 or 1. Record variable count, every equation degree, and the precise applicable Ax-Katz exponent for every restricted system, with a proof that the theorem hypotheses hold. Any applicable theorem forcing divisibility by p would exclude the known singleton; the encoding/degree calculation or claimed application must then be wrong. A bound allowing exponent zero is vacuous for deciding between 0 and 1. Verify this on complete tiny truth sets; do not infer the exact residue from the theorem alone.

Successor gate. Identify a public constructive computation of an exact 0/1 residue/receipt that costs less than enumerating the candidate domain, with arbitrary prefix restrictions preserved. That operation shares the tagged-count/conditional-coefficient bottleneck and must not be counted as independent confirmation. **Success:** exact receipt audit and calibrated counterexamples; a later algorithm needs the new residue constructor. **Failure:** one empty/nonempty receipt collision rejects existence inference, or a singleton contradicts a claimed positive divisibility bound. **Impeded:** missing exact receipt algorithm. Demonstrating divisibility is a measurement/control result, not scalar extraction or a general no-go for arithmetic counting.

L9. Probabilistic lossy projections and propagation

Prior boundary. `LEAD-EXO-lossy-object-propagation` leaves a probabilistic panel open. Exact deterministic propagation through the full group translation action on a prime-order group forces a projection to be constant or injective. This does not foreclose approximate propagation, nor do finite null observations close every probabilistic projection. The atlas's wording suggesting that one panel could close the entire probabilistic regime is too broad.

Repair the source protocol before execution. IDEA-20260817-84755b contains controls that cannot be imported as written. For uniform independent X,Y in an odd-prime subgroup including the identity, the Kummer projection `pi(R)=x(R)` with a separate infinity symbol has

`H(pi(X+Y) | pi(X),pi(Y)) = (1-1/r)^2` bits,

not exactly one bit: the conditional entropy is one when both points are nonidentity and zero otherwise. To derive this, condition on their x-coordinates. A nonidentity point has the two equally likely signs, so for nonidentity inputs the four equiprobable sign choices give the two outputs `x(X+Y)` and `x(X-Y)`, each with probability one half. They are distinct: equality would imply `X+Y = +(X-Y)` or `X+Y = -(X-Y)`, hence `2Y=0` or `2X=0`, impossible for nonidentity points in an odd-order subgroup. If either input is the identity, the output x-coordinate is determined. The probability that both are nonidentity is `(1-1/r)^2`. Conditioning both inputs to be nonidentity gives exactly one bit, but is a different sampling distribution. Its mixed curve/twist P7 domain is not a single subgroup with the operation being measured; remove it from this group-propagation statistic or give a separate domain/operation specification. The asserted universal closed-form relabeling-null formula must be derived for the declared finite-group ensemble before use; fibre sizes alone are not a supplied derivation. Monotonicity versus rho partition count is an observation to measure, not a mandatory physical law that automatically invalidates contrary data.

Exact primary panel. Measure `D(pi)=H(pi(X+Y)|pi(X),pi(Y))` by full pair enumeration on the registered 8- and 10-bit curves; select exact 12-bit enumeration or a separately validated sampler in the frozen plan. Include full point encoding (D=0), constant projection (D=0, completely lossy), Kummer x, quadratic character of x, d-th power residue labels for available d in `{3,5,7}`, top/bottom x windows `t in {1,2,4,8}`, x modulo `{3,5,7,11}`, and deterministic rho partitions `{16,32,64}`. Give infinity and zero residue explicit symbols. Omit the source's factor-base membership arm under this workstream's scope, or rename and explicitly justify a coordinate-threshold observational control without using it for relations; do not silently turn it into IC.

Exact output and null. Store each projection's image and fibre-size profile, joint frequency table or content-addressed external artifact, D, conditional candidate support sizes, and projection evaluation cost. Use a reproducible 99-element random relabeling ensemble preserving the exact fibre profile; report its spread and finite ensemble uncertainty. Independently verify D on tiny synthetic tables, especially the constant trap and exact finite-r Kummer value. Any analytic null is an additional result to prove and check against this ensemble. If a sampler is used, compare sampled and exact entropy on both smaller sizes and invalidate entropy estimates when conditional-cell occupancy makes their bias unresolved. Merely specifying ten million samples does not establish estimator adequacy.

Follow-up depth and decision. For an observed below-null gap, freeze a specific propagation/prediction algorithm and a precise depth statistic before fresh data; for example, estimate the conditional entropy of the sum of `k+1` independent points given all `k+1` projections, `k=1,...,8`, only where estimator adequacy can be demonstrated. Record all ordinary/null members and apply multiplicity-aware uncertainty across the declared panel. An exploratory gap with a held-out interval

excluding zero merits an exact structural check and a separately measured inference decoder; it does not establish scalar-bit leakage or an attack. **Failure:** one counterexample rejects claimed deterministic propagation; a sampled zero for a known lossy nonconstant member triggers exact fibre/occupancy audit. **Null:** the finite panel agrees with relabeling controls within its measured resolution. **Impeded:** invalid controls, unresolved entropy bias, no propagation algorithm, or no exact fibre data. Constant and injective projections both have D=0, so entropy alone is never the usefulness criterion.

Source anchors and current readiness

The parent source table links each source at its immutable repository revision; its selective literature and branch-coverage limits still apply.

Read-only source anchors are the four exact lead YAML rows cited above and the five full IDEA files (050, 111, 070, 156, 379), plus `ledger/proposals/IDEA-20260817-84755b.yaml`, all from the companion review's pinned `crypto-autoresearcher` revision. Readiness is conditional on registered fixtures, implementation and repository admission; no claim is made that original referenced artifact paths exist. L1 and L9 have concrete observational computations after protocol repairs. L2 and L6 have bounded diagnostic constructions but no direct locator/orientation constructor. L3's exact controls do not reopen its generic obstruction. L4, L5, L7 and L8 need explicit new constructions/theorems before a direct solver benchmark. Several original ideas are IC-specific or retired; their new direct successors must be tracked separately and credited as adaptations until novelty is independently assessed.

`dominated_by`: n/a (no attack result claimed). `sota_delta`: none measured. `novelty_status`: unverified; known machinery and source collisions are explicit. These cards create no official scientific status transition.

Development checks and limits

These are implementation self-tests on fixed tiny public synthetic inputs, not admitted scientific trials or a performance study. No solver advantage, novel identity, cryptographic-size result or independent-session verification is claimed.

Tagged reference

Sources and receipts: reference/README.md, C++20 source, fixtures, test launcher, machine-readable receipt and retained output.

Field p	Curve (a,b)	P	Curve order	Prime subgroup order r	Cofactor
5	(0,1)	(0,1)	6	3	2
17	(2,2)	(5,1)	19	19	1
23	(1,1)	(17,3)	28	7	4

The p=5 fixture is a special-j exception control. These fixtures do not represent the proposed ordinary-curve cohort. Complete point enumeration and trial division check their small orders and primes. Independently coded affine and Jacobian additions share the field reduction primitive and curve model; this is an arithmetic cross-check, with that dependency disclosed.

Each final suite reports the following measured assertion/case counts. Counts describe overlapping checks, not independent statistical samples.

Check	Recorded count
Assertions	13,731
Full-curve point-pair additions	1,181
Scalar bit inputs, including noncanonical-range inputs	44
Subgroup targets	29
Outside-subgroup on-curve targets	24
Point-transition witness candidates enumerated	58,054
One/two-step tagged boundary comparisons	2,362
Mutated traces rejected	428
Ring checks	8,225
Expected invalid-input rejections	22

Strict warning-as-error compilation, normal execution, UBSan and combined ASan/UBSan passed. LeakSanitizer failed under environment tracing; the successful combined sanitizer invocation explicitly used `ASAN_OPTIONS=detect_leaks=0`. Leak detection did not pass and remains unavailable. The initial signed-conversion compilation failure, subsequent repair, strengthened suite and sanitizer failure are retained in the receipt. No measured elapsed-time comparison was collected.

Final source SHA-256: `6694c908abf275a2be41655cbbed6f5fcfb131a0dfb300b2027a323f9e3f2685`.

Reproduce with `./reference/test.sh normal`, `./reference/test.sh ubsan` and `ASAN_OPTIONS=detect_leaks=0 ./reference/test.sh asan-ubsan`. The launcher builds in a disposable temporary directory; its only target interface is `--self-test`. The fixture-discovery utility is provenance scaffolding for these tiny fixtures, not the scientific-cohort generator.

Fixed-window control

window-control.md derives the exact binary rank r−1 and the tagged row/column margins of the full point-state window matrix. The written argument applies to odd r>1. It is a known scalar-window control and supplies no unknown target's scalar labels.

The C++ regression checks r in {3,5,7,11,13,17} and d in {1,2,3,4}: 24 cases, 24 ranks, 448 tagged margins, 24 deliberate mutations and 18 rank calibrations. Both normal and ASan/UBSan execution passed with leak checking explicitly disabled for the environment limitation. Its source SHA-256 is `d7948445e4c7cf75e6c90417107886e935110aee4d9c903a8581b59054221a94`.

The rank is over F_2, not the square-zero ring. High rank of this full matrix only bounds exact rank-one decompositions of that flattening. It does not bound arbitrary bit-level algorithms or target-conditioned evaluation time; full rank and cheap implicit computation can coexist.

What is still missing

The reference has deterministic **point-level** states. It does not compile modular arithmetic into Boolean gates/carries, establish unique wire extensions for such a circuit, search a frozen rewrite grammar, or evaluate a new cancellation construction. Fixed-window agreement is a known positive control. No discovery/confirmation cohort, blinded labels, rho comparison, formal proof assistant build or admitted experiment has run.

The methodology review is same-session assistance with access to the plan. It is useful criticism but is not a formal independent review, blind re-derivation or novelty certificate. The source corpus, source hashes and earlier literature coverage limits remain those in SOURCES.md.

The lossy-panel correction in LEADS.md is a derived finite-group reference, not a new measured result or correction to frozen evidence. It requires separate checking before an official source/status update.